

Are Lexical Representations Separable? Item and Category in Learned Distributed Representations

Anonymous ACL submission

Abstract

PLACEHOLDER — author to write.

1 Introduction

2 Experiment 1: What a Learner Builds from a Known Language

The goal of experiment 1 is to validate our measurement on an artificial language whose structure we control as well as examine how a learner represents that structure.

2.1 Dataset

In order to examine the representation of linguistic concepts, we constructed an artificial language which comprises a probability distribution over a vocabulary of 600 tokens. Every observation is a triple: a **form** (one of 60), a **class** (one of 4), and a **context** drawn from a pool of 400. The motivation for this setup is to give each word a naturalistic environment: the word occurs in a context that favors a specific class-membership (e.g., *walk* as a verb occurs in a different context than *walk* as a noun). Each form is observed 24 times at each class level, every time in a different context, which is what a word in a corpus looks like: one type, many occurrences, each in different company. Consider *walk* used as a verb, in one of those contexts. Four things determine the distribution over the next token. *Walk* itself contributes the same amount whichever class it is used as, and the verb class the same amount for every verb. The third contributes whatever is specific to *walk*-as-a-verb over and above those two: every form and every class has its own random vector, and this term multiplies *walk*'s by the verb's, entry by entry. A sum could be pulled back apart into a form part and a class part; a product cannot, which is what makes this term genuinely joint. The context contributes the rest, which is noise as far as our question goes, though the model

still has to encode it exactly as a language model has to encode the rest of a sentence. The four are weighted, summed and passed through a softmax:

$$P(\text{next token} \mid f, c, x) = \text{softmax} \left(s \left[w_{\alpha} \text{item}(f) + w_{\beta} \text{class}(c) + w_{\gamma} \text{int}(f, c) + w_{\text{ctx}} \text{context}(x) \right] \right) \quad (1)$$

Each term is scaled to unit length before it is weighted, so changing the weights changes the mix of the three effects without changing how much signal there is in total, and the task stays equally hard everywhere in the grid.

The weights w_{α} , w_{β} and w_{γ} in Equation 1 control how much each of the three effects contributes to the true distribution, and we cross them factorially over $\{0, 0.5, 1, 2\}$: zero removes an effect from the language altogether, and the other three values make an effect half as strong as another, equal to it, or twice it. That gives $4^3 = 64$ combinations, and dropping the one in which all three are zero leaves 63. Fifteen of those plant no item effect, fifteen no class effect and fifteen no interaction, so the measure is asked to report zero in every case where zero is the truth. Similarly, w_{ctx} governs how much the context contributes, so we cross those 63 languages with three context weights, ranging from context being a minor part of the true distribution to the dominant one. We train a model on each of the resulting languages at hidden widths of 8, 16, 32, 128, with either a linear or a ReLU activation (Section 2.2). The whole grid is then run with five random seeds, each redrawing the language and reinitializing the model, giving 7,560 runs in all, of which 99.7% trained to a loss plateau.

It is worth noting that entanglement is guaranteed under a specific subset of the languages. That is, the structure needs 16 dimensions, so its ratio to the hidden width, which we call **capacity**, runs

from 0.125 to 2.0. Below one there is room to give every component its own axis, so any entanglement was chosen; above one there are more components than dimensions to put them in, so they cannot all have their own axis and entanglement is forced (Elhage et al., 2022).

2.2 Method

2.2.1 The learner

The input to the learner model comprises three pieces of information: which form, which class, and which context. Each has its own embedding, and the three are concatenated and passed to a standard multi-layer perceptron (MLP) with a single hidden layer of width d and either a linear or a ReLU activation function, which outputs a distribution over the vocabulary (Figure 1). The training objective is thus to minimize cross-entropy between the model’s predicted distribution over the vocabulary and the true distribution over the vocabulary.

2.2.2 Decomposing a representation

We measure separability by decomposing a word-in-context representation into its lexical representation, its class representation, and its idiosyncratic representation. The representation is the MLP’s hidden layer, which for a given form and class varies across the contexts it is observed in, so let $M[i, c] \in \mathbb{R}^d$ be the mean of those hidden states for item i used at class level c . We decompose that grid by centering, so that

$$M[i, c] = \mu + \alpha(i) + \beta(c) + \gamma(i, c), \quad (2)$$

where μ is the grand mean; $\alpha(i)$, item i ’s mean across class levels minus μ , carries the lexical information; $\beta(c)$, the mean across items at class level c minus μ , carries the class information; and $\gamma(i, c)$ carries the idiosyncratic information, what item i at class level c holds that neither marginal accounts for. Because every item is observed at every class level with equal weight, neither marginal absorbs part of the other. The four terms add back to $M[i, c]$ exactly, and no two of them vary together across the grid, so their squared lengths sum to the total and we report each as a share of it.

Each of these averages is taken over finitely many observations, which differ only in context, so each carries some of the context term of Equation 1 as noise. For γ this matters: a squared length is

always positive, so its size would come out above zero even for a language with no idiosyncratic information in it at all. We therefore split the observations behind each average in half, run the decomposition on each half separately, and take the dot product of the two estimates of γ , $\langle \gamma_A, \gamma_B \rangle$, which is large when the halves agree about the idiosyncratic information and near zero when they do not. Noise that differs between the halves cancels, and unlike a squared length this can come out negative, which is what allows a test: significance comes from permuting which item of one half is paired with which item of the other, preserving each component’s size while destroying the correspondence between them.

For each representation we report two measurements. The first measures the fraction of the representation accounted for by lexical, class and idiosyncratic information: each component’s squared length, summed over its vectors and divided by the total, so that the three fractions sum to one. The second measures whether those components occupy the same directions. In order to do so, we project the item vectors onto the class direction and record the fraction of their squared length that survives, and we project the interaction onto the smallest subspace containing both the item and the class vectors (i.e., the span of the two subspaces). Each fraction is read against r/d , what a randomly oriented effect would place in a subspace of rank r at width d . A representation is separable if a word’s representation is the sum of an item part and a class part with the two on their own axes, so it fails either when the components are non-orthogonal, so that projecting out the class also removes item information, or when there is a third component that is neither an item part nor a class part, whatever its orientation. The question is accordingly not whether class information is present, but how it is carried: on a single direction shared across words, or on a separate direction for each of them.

2.3 Results

The weights in Equation 1 describe the true distribution the model is trained to match, while the measure reads the representation it builds, so their numbers do not match: a learner handed a language that is 85% interaction can build a representation in which the interaction is a far smaller share, because the representation only has to be enough to produce the answer. Validation asks that the measure track what was planted, read zero when zero

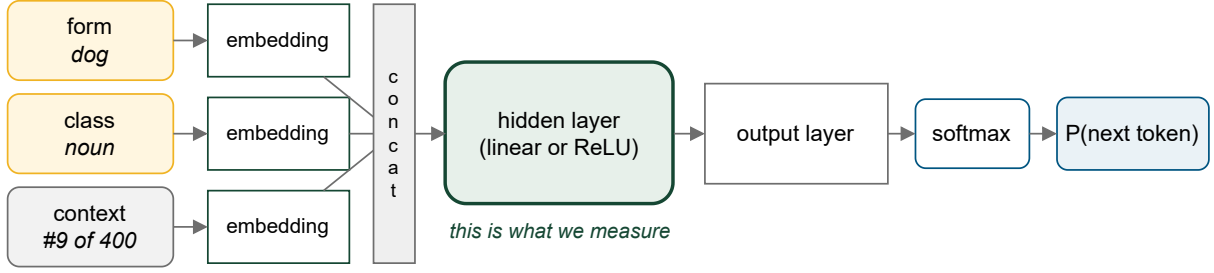


Figure 1: What the learner sees. Every form is observed 24 times at each class, each time in a different context; that variation across contexts is what the measurement later splits in half.

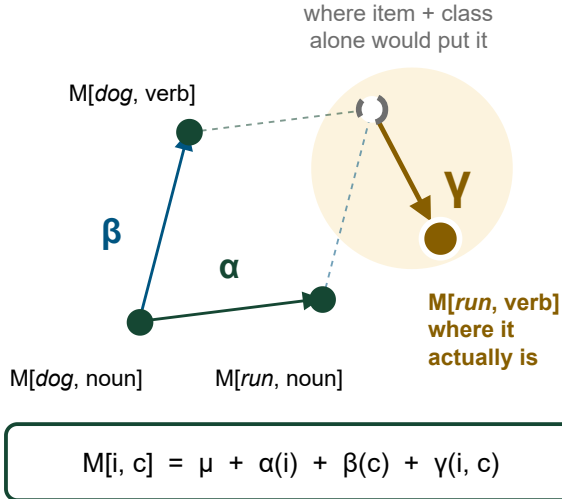


Figure 2: If a word’s representation were an item part plus a class part, three corners of a 2×2 grid of means would fix the fourth and the parallelogram would close. It does not. γ is how far the actual mean sits from where that additive account puts it, and it is not error: the four components exhaust the grid exactly, so γ is the part that belongs to the pairing of a particular word with a particular class. The arrow’s length is $\|\gamma(i, c)\|$, and with two items and two classes it is the whole interaction, since γ then has a single degree of freedom and the four means share one vector up to sign. What we report is each component’s squared length as a share of the structure across the grid.

was planted, and have a known floor. Appendix B reports the separate check in which the components are planted directly in a representation.

The measure reports zero when the answer is zero. Under the linear activation the hidden layer is a weighted sum of the three embeddings, so its interaction is identically zero however much the language contains (Appendix A), and every reading the measure gives there is its own noise. Across all 2,880 runs where the language *did* contain an interaction, it reports a median share of 0.0010 and flags one in 5.0% of them, the nominal rate for a test at the 5% level; where no interaction was planted it flags 4.9%. This calibration comes from a condition whose true value is provably zero, measured at every width, sample size and noise level in the grid.

The measure also tracks what is there. Under ReLU the measured interaction rises monotonically with the planted one (Figure 3, left), and the rate at which it is flagged rises with it, from 38% at the smallest nonzero setting to 96% at the largest. With no interaction planted, ReLU flags 2.9% of runs, below nominal: the nonlinearity does not manufacture an interaction out of an additive language. The measured shares are smaller than the planted ones throughout, which is what that distinction predicts; what matters is that the relationship is monotone and that zero maps to zero.

Item and class effects entangle only when there is no room for them not to, and this is the substantive result (Figure 3, right; Table 1). With capacity well below one, ReLU’s item-class overlap is 0.0092 against a chance level of 0.0156, so the two effects are *more* orthogonal than two arbitrary directions would be, which is what a learner that has room and uses it looks like. At capacity 1 the overlap is 0.402 against a floor of 0.0625, about 6.4 times chance, and at capacity 2 it is 0.530; the linear condition stays at or below chance throughout. This

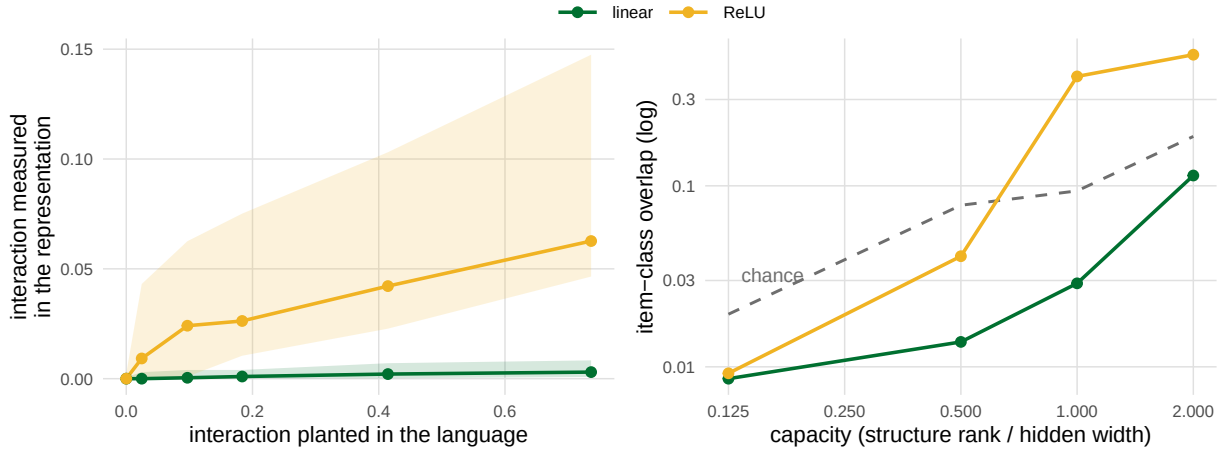


Figure 3: Experiment 1, all 7,560 runs. Left: the measured interaction against the amount planted in the language, with interquartile ranges. The linear condition cannot represent an interaction whatever the language contains, and reads zero throughout; the ReLU condition tracks what was planted. Right: overlap between the item and class effects against capacity, with the dashed line marking what a randomly oriented effect would give at that width. Below capacity the two effects are more orthogonal than chance; once dimension counting makes separation impossible they are several times chance. Note the log axis, and that the floor moves by a factor of eight across these widths.

condition	capacity	interaction	overlap	chance	ratio
linear	0.125	0.005	0.0086	0.0234	0.37
linear	0.500	0.002	0.0137	0.0938	0.15
linear	1.000	0.000	0.0289	0.1250	0.23
linear	2.000	0.000	0.1140	0.2500	0.46
ReLU	0.125	0.080	0.0092	0.0156	0.59
ReLU	0.500	0.035	0.0408	0.0625	0.65
ReLU	1.000	0.007	0.4022	0.0625	6.44
ReLU	2.000	0.001	0.5303	0.1250	4.24

Table 1: Experiment 1, by capacity, over runs where all three effects were planted. Chance is the share of a randomly oriented effect that would fall in the class direction at that width, and the ratio is the observed overlap against it. Below one, a fully separable solution is reachable by dimension counting; above one it is not.

is superposition in a setting where we can watch it happen: when the number of directions the structure needs exceeds the number available, the components stop being orthogonal because they cannot all be, and nothing about the language changed across those runs.

The measure also works when the representation primarily encodes contextual information unrelated to the specific form and class. When the context term accounts for more than 80% of the variance in the logits, leaving under a fifth for the form, the class and their interaction combined, ReLU’s measured interaction holds at 0.021 against 0.038 when context is minor, and the flag rate falls from 70% to 44%. Power degrades, as it must with twenty-four observations per item and level, but

the estimate does not drift. That is the regime Experiment 2 sits in, since a word’s hidden state varies mostly with the sentence it appears in rather than with the word or its class.

Two things carry into Experiment 2. The instrument reads zero when the answer is zero and tracks the answer when it is not, at the sample size and noise level the language models will be measured at. And whether the item and class effects come out orthogonal is governed by whether the representation had room, which is a fact about capacity rather than about the language, so near-orthogonality in a wide model is uninformative on its own and has to be read against its own chance level.

3 Experiment 2: Item and Category in Language Models

Experiment 1 established that a learner fitting an interactive language cannot represent it as item plus class, and that a nonlinearity additionally makes the components non-orthogonal. Experiment 2 applies the same decomposition to the contextual representations of pretrained language models, across three linguistic distinctions chosen to span grammar and meaning.

3.1 Dataset

For part of speech and grammatical role we concatenate six general-domain English Universal Dependencies treebanks (Nivre et al., 2020): EWT, GUM, LinES, ParTUT, GENTLE and PUD, giving

distinction	levels	analyzed	tokens
part of speech	NOUN / VERB	78	4,722
grammatical role	<i>nsubj</i> / <i>obj</i>	49	2,150
metaphor	literal / figurative	NA	NA

Table 2: The three constructions. All three key the item on the **surface form**, so the token is identical at both levels of the distinction. *Analyzed* counts the forms attested often enough at each level to estimate a mean at each.

41,095 sentences and 706,323 tokens. A lemma must be attested at both class levels often enough to estimate a mean at each, and a single treebank yields too few such lemmas. For metaphor we use the VUA20 corpus (Leong et al., 2020; Steen et al., 2010), restricted to content words, which supplies NA labeled tokens.

All three constructions are **same-token**: the item’s surface form is identical across the two class levels, so the representation is read from the same token in both, and no part of any difference can come from tokenization.

Keying the item on the *lemma* instead admits more items (130 rather than 78, drawn from 999 lemmas attested as both), but a lemma pools every inflected form: *run* and *runs* as a noun against *run*, *runs*, *ran* and *running* as a verb. Those are different tokens, and the consequence shows up at the embedding layer, which has no context to work with and so can only distinguish levels that differ in their input. Keyed on the lemma the layer-0 interaction is 0.083 to 0.172 across models; keyed on the form it is 0.0010 to 0.0028. The difference is inflection rather than representation, and keying on the form removes it at the cost of a smaller item set. Appendix C shows what the measure does on a distinction where the property cannot be arranged at all.

Identifiability. The interaction $\gamma(i, c)$ is estimable only for an item observed at more than one level, since it is defined by how an item’s representation shifts across levels, and we additionally require at least 10 tokens for each item at each level so that a mean can be estimated and then split in half. This restricts the scope for part of speech, where it selects forms genuinely used both ways, such as *run* or *hit*. Role and metaphor carry no such restriction, since any noun occurs as both subject and object and any content word occurs both literally and figuratively, which is one reason they broaden the claim. The median item holds around

24 tokens at each level across the three constructions, the sample size the measure is validated at in Appendix B.

We measure two model families at three matched scales: OPT-BabyLM, trained on developmentally plausible quantities of child-available language (Warstadt et al., 2023), and Pythia, trained on the Pile (Biderman et al., 2023), at 125M/160M, 350M/410M and 1.3B/1.4B parameters, with widths of 768, 1024 and 2048. Appendix D gives parameter counts and training-token estimates.

3.2 Method

The decomposition, the two reported quantities and the replication logic are those of Section 2.2. For every token we take the hidden state of the word’s **last subword** at every layer, from the embeddings through the final block, aligning subwords to corpus tokens with the tokenizer’s word alignment. Sentences are drawn in a fixed random order and processed until roughly 300K tokens are extracted per model, so all models see the same material, and representations are extracted once with every analysis running offline from them.

The two independent estimates come from the corpus itself: each item at a level is a word used across many sentences, so we split its token contexts into two disjoint halves and estimate the grid from each. What cancels is context noise, the variation in a word’s hidden state across the sentences it appears in, which is the same thing the split cancels in Experiment 1.

Standardization. Transformer hidden states contain a few *rogue* dimensions whose variance exceeds the rest by two to three orders of magnitude (Timkey and van Schijndel, 2021), and these would otherwise dominate any subspace estimated from the raw scatter, so we divide each dimension by its standard deviation before decomposing. This is a diagonal rescaling rather than a whitening, so it needs no favorable ratio of samples to dimensions; the measure is accordingly invariant to per-dimension rescaling but not to arbitrary linear reparameterization, which we return to in Appendix A.

3.3 Results

The two main effects sit on separate axes. At the deepest layer, item-class overlap runs 0.0032 to 0.0067 for part of speech and 0.0025 to 0.0062 for grammatical role, so projecting out the class direction leaves better than 99% of the item effect

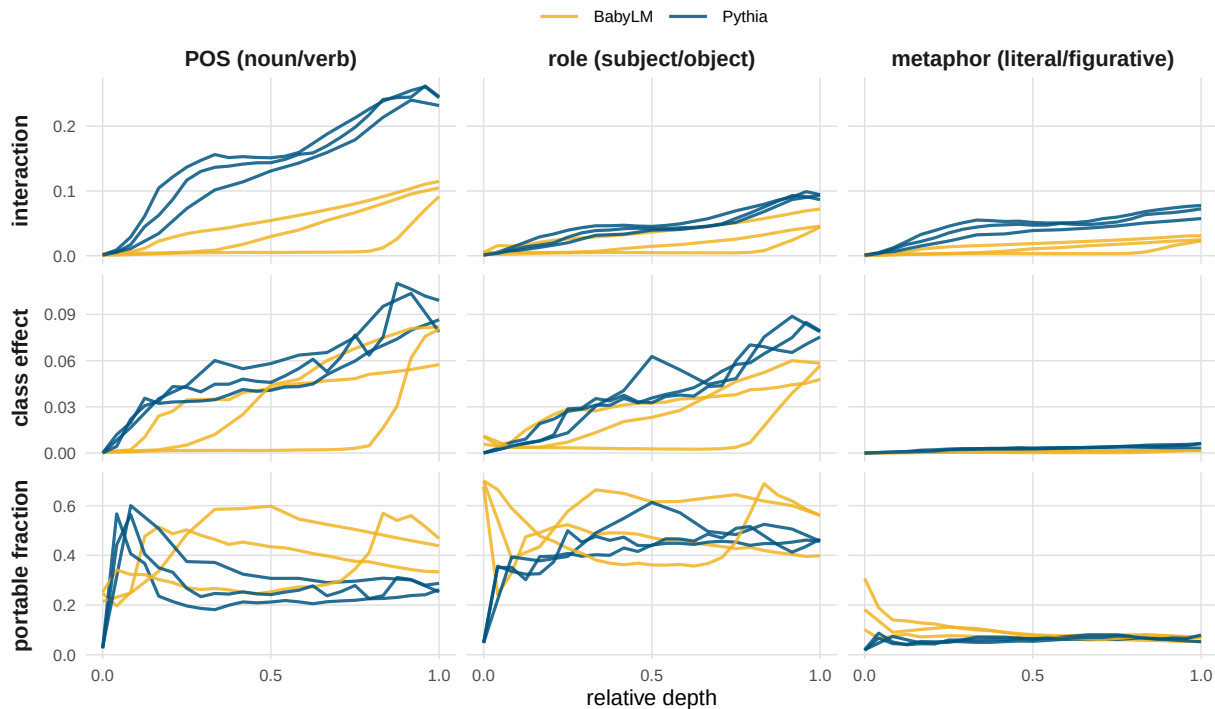


Figure 4: Experiment 2, by relative depth (0 = static embeddings, 1 = final block). Top: the interaction’s share of representational structure. Middle: the class effect’s share. Bottom: the portable fraction, the class effect’s share of the two together, which is the quantity the three distinctions are ordered by. Each line is one model; color distinguishes the two training corpora. Both components are built by depth, and the interaction is the larger of the two throughout. The deepest-layer values, with their chance levels, are in Table 9.

intact. Those values are two to seven times what a randomly oriented effect would give at these widths (Appendix B), so they are not exactly zero, but they are small enough that these representations do not achieve class structure by borrowing the directions that carry word identity. Where a shared class direction exists, it is its own direction.

The interaction, however, does not have axes of its own, and this is what decides whether a third component could simply be added to the description and separability recovered in a wider sense. For part of speech 0.191 of the interaction lies in the span of the two main effects, for role 0.131 and for metaphor 0.227, against chance levels of 0.076, 0.048 and 0.176 (Table 9). The partition of the grid says nothing about direction, so these numbers are not implied by the decomposition: the item-by-class term reuses the very directions that carry item identity and class membership. Read with the first overlap, they say something more specific than “not separable”. The failure is not that item and class have been collapsed onto shared axes, since those remain distinct; it is that the part of class structure specific to each word is written in the same directions as everything else. For metaphor

the question does not arise, because the class effect never rises above the level at which an orientation is worth reporting, which is itself the first indication of how little of metaphoricity is carried by a shared direction.

A substantial interaction is there, and depth builds it. In all three distinctions the interaction is essentially absent at layer 0, between 0.0002 and 0.0051 across every construction and model, and rises monotonically with depth to between 0.023 and 0.246 (Figure 4, top row). Because the item is keyed on the surface form, the token is identical at both class levels, so the static embedding layer has nothing to distinguish them with, and it does not. Everything the measure finds is therefore created by contextualization rather than inherited from the input, and it is not positional: zeroing OPT-BabyLM’s absolute position embeddings leaves the class structure more word-specific rather than less (Appendix F). This is the component that breaks separability, since it is neither a fact about the word nor a fact about the class.

The class effect grows with depth as well (Figure 4, middle row), and that matters for the framing. Models are not choosing between abstraction

and entanglement; contextualization builds both at once. A description of these representations as either abstract or exemplar-like is therefore underdetermined by the data, and the useful question is not which one they are but what the balance is.

That balance is graded across distinctions. Table 9 reports it as the *portable fraction*: of the structure that distinguishes the two class levels, how much is a direction shared across words rather than specific to each word. Grammatical role is the most portable at 0.46, part of speech next at 0.31, and metaphor by far the least at 0.07, so only about a fourteenth of metaphor’s class-relevant structure is carried by a shared direction. The ordering is worth noting: part of speech has the largest class effect of the three in absolute terms, but also much the largest interaction, so proportionally less of what distinguishes its levels is portable than for grammatical role. Metaphoricity is close to being encoded word by word, and what makes a use of *plant* figurative is largely not what makes a use of *course* figurative.

4 Experiment 3: Does the Interaction Carry the Class?

Experiment 2 shows that the interaction is large and that it replicates across independent halves of a word’s occurrences, but neither shows what it is *about*. An item-by-class term captures any reliable difference between a word’s uses at the two levels, and some of that difference is context correlated with the label rather than the label itself, since metaphorical uses occur in different sentences from literal ones and subjects appear in different constructions from objects. So γ is an upper bound on class-binding, and the question is whether any of it is the class.

4.1 Method

We test it by decoding. If γ encodes the class, the class should be recoverable from the part of a token’s representation that is neither the item effect nor the shared class direction, and we remove both explicitly. All decoding happens *within* a single word, so we ask whether *plant*’s figurative uses can be told from *plant*’s literal ones, never whether *plant* can be told from *account*; the item effect is constant within a word and so cannot contribute. We then estimate β as the difference between the two class means pooled over all items, normalize it, and remove that component from every token,

distinction	family	words	class	null
part of speech	BabyLM	78	0.93 [0.91, 0.94]	0.52 [0.49, 0.55]
part of speech	Pythia	78	0.94 [0.93, 0.95]	0.50 [0.48, 0.53]
grammatical role	BabyLM	49	0.85 [0.82, 0.87]	0.50 [0.47, 0.53]
grammatical role	Pythia	49	0.84 [0.82, 0.87]	0.52 [0.49, 0.55]
metaphor	BabyLM	180	0.65 [0.63, 0.66]	0.49 [0.47, 0.50]
metaphor	Pythia	180	0.70 [0.68, 0.71]	0.49 [0.48, 0.51]

Table 3: Class decoded from the interaction, at the deepest layer of the largest model in each family. The item is held fixed and the shared class direction is projected out, so neither the item effect nor the class effect can contribute. *Class* is the accuracy of that decode and *null* permutes class labels within each word; brackets are percentile bootstrap intervals over words. Chance is 0.50.

so that whatever separates the classes afterwards is not the direction they share.

What remains is the interaction: the part of a word-at-a-level that is neither a fact about the word nor a fact about the level. We decode it with a cross-validated logistic probe, five folds, classes balanced by subsampling within each word so that chance is 0.50, and average the resulting accuracies over words, with a word contributing only if it has at least eight tokens at each level. Accuracies are averaged unweighted over words and some words rest on few tokens, so the mean carries a percentile bootstrap interval over words. The whole procedure is then repeated with the class labels permuted within each word, which preserves every word’s token count and class balance while destroying the labelling, and that null is where a probe reading its own cross-validation folds would land. We run this at the deepest layer of the largest model in each family, for all three constructions.

4.2 Results

The interaction is about the class. In every construction and both families the class is recoverable from the interaction at well above chance while the permuted null sits on chance (Table 3), reaching 0.94 for part of speech, 0.84 for grammatical role and 0.70 for metaphor, with bootstrap intervals excluding the nulls by a wide margin in all six cases. The null landing on chance is what makes those accuracies interpretable, since it is the same probe, the same folds, the same word-level averaging and the same small per-word samples with only the labelling destroyed.

The effect is graded the same way the interaction’s size is. Part of speech decodes best, role next, metaphor least, which is the ordering of the

portable fraction in Table 9 read backwards: the distinctions whose class-relevant structure is least portable are also the ones whose interaction carries the class least strongly. Two measurements, one geometric and one functional, agree on the same ranking.

Metaphor is the case where this matters most. Its class effect never clears the threshold at which an orientation is worth reporting (Section 3.3), so on the geometric evidence alone one could conclude that metaphoricity is not represented at all, yet it decodes from the interaction at 0.70 against a null of 0.49. It is represented; it is simply not represented as a direction shared across words, and a method that looks only for a shared direction would have missed it.

This result has a definite scope. It shows that γ carries the specific label rather than generic correlated context, since generic context would not survive holding the word fixed and removing the shared axis. It does not show that γ is *only* the label: context correlated with the class within a word survives both removals, so the interaction remains an upper bound on class-binding, and this raises the floor rather than closing the gap.

5 Discussion

Limitations

Everything here is English, and rests on three distinctions measured in two model families at three scales each. The graded ordering we report is an ordering across *these* distinctions, and whether it holds for aspect, animacy or definiteness is untested.

The measure is invariant to per-dimension rescaling and to rotation, but not to arbitrary linear reparameterization, so overlap between the item and class effects is relative to the model’s own basis. We take that to be the basis of interest, since LayerNorm and the feed-forward nonlinearities act coordinate-wise and a downstream output layer operates on the unwhitened space; a reader who wants a fully gauge-free quantity will not find one here. The component sizes, their growth with depth and their significance do not depend on that choice.

Two things we would like to have done. These are endpoint measurements, so they say nothing about the order in which the two components are learned, which needs training checkpoints. And the interaction captures any systematic within-word difference between the levels, so a distinction re-

alized through the contexts a word appears in is measured together with those contexts. For a contextual model we take that to be what representing the distinction consists in, but separating the two would need a construction in which they can be varied independently.

References

- Stella Biderman, Hailey Schoelkopf, Quentin Anthony, Herbie Bradley, Kyle O’Brien, Eric Hallahan, Mohammad Aflah Khan, Shivanshu Purohit, USVSN Sai Prashanth, Edward Raff, Aviya Skowron, Lintang Sutawika, and Oskar van der Wal. 2023. Pythia: A suite for analyzing large language models across training and scaling. In *Proceedings of the 40th International Conference on Machine Learning (ICML)*.
- Nelson Elhage, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, and 1 others. 2022. Toy models of superposition. *Transformer Circuits Thread*.
- Chee Wee (Ben) Leong, Beata Beigman Klebanov, Chris Hamill, Egon Stemle, Rutuja Ubale, and Xinyang Chen. 2020. A report on the 2020 VUA and TOEFL metaphor detection shared task. In *Proceedings of the Second Workshop on Figurative Language Processing*.
- Joakim Nivre, Marie-Catherine de Marneffe, Filip Ginter, Jan Hajič, Christopher D. Manning, Sampo Pyysalo, Sebastian Schuster, Francis Tyers, and Daniel Zeman. 2020. Universal dependencies v2: An evergrowing multilingual treebank collection. In *Proceedings of the 12th Language Resources and Evaluation Conference (LREC)*.
- Gerard J. Steen, Aletta G. Dorst, J. Berenike Herrmann, Anna A. Kaal, Tina Krennmayr, and Trijntje Pasma. 2010. *A Method for Linguistic Metaphor Identification: From MIP to MIPVU*. John Benjamins.
- William Timkey and Marten van Schijndel. 2021. All bark and no bite: Rogue dimensions in transformer language models obscure representational quality. In *Proceedings of the 2021 Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- Alex Warstadt, Aaron Mueller, Leshem Choshen, Ethan Wilcox, Chengxu Zhuang, Juan Ciro, Rafael Mosquera, Bhargavi Paranjabe, Adina Williams, Tal Linzen, and Ryan Cotterell. 2023. Findings of the BabyLM challenge: Sample-efficient pretraining on developmentally plausible corpora. In *Proceedings of the BabyLM Challenge at CoNLL*.

A The Decomposition and Its Estimator

A.1 The decomposition as a projection

For items $i \in \{1, \dots, I\}$ crossed with class levels $c \in \{1, \dots, C\}$, let $M[i, c] \in \mathbb{R}^d$ be the mean representation of item i at level c . Regard the grid as an element of $\mathbb{R}^I \otimes \mathbb{R}^C \otimes \mathbb{R}^d$ and define averaging and centering projectors on the first two factors:

$$P_I = \frac{1}{I} \mathbf{1}_I \mathbf{1}_I^\top, \quad Q_I = \text{Id}_I - P_I, \quad P_C = \frac{1}{C} \mathbf{1}_C \mathbf{1}_C^\top, \quad Q_C = \text{Id}_C - P_C.$$

The four components of Equation 2 are the four tensor-product projections of M :

$$\mu = (P_I \otimes P_C)M, \quad \alpha = (Q_I \otimes P_C)M, \quad \beta = (P_I \otimes Q_C)M, \quad \gamma = (Q_I \otimes Q_C)M,$$

which sum to M because $(P_I + Q_I) \otimes (P_C + Q_C) = \text{Id}$. They are mutually orthogonal idempotents, so this is an orthogonal direct sum with dimensions 1, $I-1$, $C-1$ and $(I-1)(C-1)$. This is a geometric construction rather than a statistical model: there is no error stratum, no F -ratio and no distributional assumption.

Two consequences are used in the body. First, the side constraints $\sum_i \alpha_i = 0$, $\sum_c \beta_c = 0$ and $\sum_i \gamma_{ic} = \sum_c \gamma_{ic} = 0$ follow from the projectors rather than being imposed. Second, an **additive** grid, one of the form $M[i, c] = A(i) + B(c)$, has $\gamma = 0$ *identically*: writing $\bar{A}$ and $\bar{B}$ for the averages, $\mu = \bar{A} + \bar{B}$, $\alpha_i = A(i) - \bar{A}$, $\beta_c = B(c) - \bar{B}$, and the remainder cancels exactly. This is what makes the linear condition of Experiment 1 a negative control rather than an assumption.

A.2 Sizes

Orthogonality gives an exact partition of the total squared length across the grid,

$$\sum_{i,c} \|M[i, c] - \mu\|^2 = C \sum_i \|\alpha_i\|^2 + I \sum_c \|\beta_c\|^2 + \sum_{i,c} \|\gamma_{ic}\|^2,$$

with C and I appearing as replication counts: α_i is shared across the C levels and β_c across the I items. Sizes are reported as shares of the total. All three terms scale linearly in I , so the shares do not depend on how many items the grid contains and are comparable across constructions of different size.

A.3 Item-class overlap and its chance level

Subspaces are orthonormal bases from the leading right singular vectors, with rank set by the participation ratio $r = \text{round}((\sum_k \lambda_k)^2 / \sum_k \lambda_k^2)$ of the scatter eigenvalues, capped at the algebraic maxima above. With two class levels β spans one direction, so $r_{\text{class}} = 1$ throughout Experiment 2.

Overlap is the fraction of squared length inside a subspace, $\|VS\|_F^2 / \|V\|_F^2$. Its chance level is not zero: for a randomly oriented V ,

$$\mathbb{E}[\|VS\|_F^2] = \text{tr}(V^\top V) / d \cdot r \implies \mathbb{E}[\text{overlap}] = r/d,$$

whatever V 's rank or shape. At $d = 2048$ that is 0.0005, and Appendix B measures it directly at 0.0010. Any overlap has to be read against that number, which moves by two orders of magnitude across the widths used in this paper.

A.4 Why the interaction is estimated as a cross-product

The grid means are estimated from finitely many observations, $\hat{M} = M + E$ with $\mathbb{E}[E] = 0$. The decomposition is linear, so $\hat{\gamma} = \gamma + (Q_I \otimes Q_C)E$ and

$$\mathbb{E}[\|\hat{\gamma}\|^2] = \|\gamma\|^2 + (I-1)(C-1) d \sigma^2.$$

A squared length is therefore bounded below by its own noise floor and cannot read zero even when the true interaction is zero. Permutation does not repair this: a squared projection norm is a magnitude,

its expectation under the null is that same floor, and the permuted replicates carry the floor too, so a real interaction and noise of the same size give the same statistic.

Given two independent estimates $\hat{\gamma}_A = \gamma + G_A$ and $\hat{\gamma}_B = \gamma + G_B$ with $G_A \perp\!\!\!\perp G_B$,

$$\mathbb{E}\langle \hat{\gamma}_A, \hat{\gamma}_B \rangle = \|\gamma\|^2 + \langle \gamma, \mathbb{E}G_B \rangle + \langle \mathbb{E}G_A, \gamma \rangle + \langle \mathbb{E}G_A, \mathbb{E}G_B \rangle = \|\gamma\|^2,$$

exactly unbiased at any noise level, and able to come out negative, which is what makes > 0 a test rather than a threshold on a positively biased quantity. Significance permutes the item index of one estimate against the other, preserving every magnitude while destroying the correspondence, over 200 draws at the 5% level.

The same construction is applied to the item effect. The **class effect is not cross-denoised**: it is reported as $I \sum_c \|\hat{\beta}_c\|^2$ from the full grid, on the grounds that $\hat{\beta}_c$ averages over all I items and is well estimated. That addresses variance rather than bias, and Appendix B measures the floor it leaves.

A.5 Invariance

The measure is exactly invariant to per-dimension rescaling, which the standardization absorbs, and the functionals are invariant to orthogonal rotation of already-standardized data. It is **not** invariant to general linear reparameterization: we do not whiten. Whitening would make a linear representation read separable in any basis, but it requires estimating a $d \times d$ covariance and it discards the basis the model actually computes in. LayerNorm and the feed-forward nonlinearities act coordinate-wise, and a downstream output layer operates on the unwhitened space, so the question of interest is whether item and class occupy separate axes *in the model's own basis*. Overlap values are therefore basis-relative, while the sizes, their growth with depth and their cross-denoised significance do not depend on the choice.

B Validating the Measure

Experiment 1 plants structure in a *language* and measures a trained model's representation. A discrepancy there is ambiguous between the measure missing something and the model not mirroring the language, and the second is known to happen: the linear condition cannot represent an interaction at all. This appendix removes the ambiguity by planting μ , α , β and γ **directly in a representation**, with no model and no training, and passing the result to the same function the language models are measured with. Ground truth now sits at the level the measure operates on.

Construction. The three components are drawn at random and then centered so each satisfies the constraints its role implies: α and β sum to zero over their own index, γ over both. Without that a planted γ would partly be an item or class main effect and the planted shares would be wrong before the measure saw them. Each is rescaled to hit a target share of the total squared length across the grid, under the same weighting the measure uses. Each mean is then observed n times under isotropic noise, whose scale is set as a multiple of the mean squared length. The grid crosses the planted shares (the same factorial Experiment 1 runs, mapped into share space, plus the three compositions Experiment 2 measures), item counts from 27 to 180, two and four class levels, widths from 8 to 2048, 10, 24 and 100 observations per item and level, three noise levels, planted-orthogonal and planted-non-orthogonal effects, and five seeds: 252,000 runs.

Recovery. At Experiment 2's regime, 24 observations per item and level in 768 or more dimensions, the measured interaction matches the planted one across the whole range (Table 4), and is flagged in essentially every run where anything was planted and none where nothing was.

planted	runs	measured	flagged
0.000	6300	0.0000 [0.0000, 0.0074]	0%
0.030	900	0.0323 [0.0244, 0.0372]	98%
0.048	1800	0.0494 [0.0413, 0.0680]	100%
0.056	1800	0.0571 [0.0493, 0.0907]	100%
0.059	1800	0.0605 [0.0526, 0.1024]	100%
0.100	1800	0.1006 [0.0921, 0.1051]	100%
0.111	900	0.1149 [0.1035, 0.1274]	100%

(continued)

planted	runs	measured	flagged
0.167	1800	0.1701 [0.1582, 0.2188]	100%
0.190	1800	0.1926 [0.1817, 0.2709]	100%
0.200	1800	0.2020 [0.1910, 0.2950]	100%
0.240	900	0.2407 [0.2284, 0.2445]	100%
0.333	900	0.3372 [0.3203, 0.3603]	100%

Table 4: Recovery at 24 observations per item and level in 768 or more dimensions, pooled over noise levels, item counts and widths. Brackets are the 5th and 95th percentiles across runs.

Floors. The interaction and the item effect read essentially zero when nothing is planted, at every sample size. The class effect does not, and this is the one quantity in the paper that carries a floor worth knowing (Table 5): at 24 observations per item and level a true zero reads 0.0036. This is the price of estimating the class effect as a squared length rather than a cross-product, and it is why the metaphor result in Section 3.3 is stated as an inability to distinguish the class effect from zero rather than as a measurement of something small. Metaphor’s observed class effect, 0.0016 to 0.0062 at the deepest layer, sits at that floor.

effect	observations	median	95th pct	flagged
class effect	10	0.00836	0.0849	—
class effect	24	0.00359	0.0365	—
class effect	100	0.00085	0.0093	—
interaction	10	0.00000	0.0557	2.6%
interaction	24	0.00002	0.0312	1.4%
interaction	100	0.00003	0.0084	0.4%
item effect	10	0.00008	0.0559	—
item effect	24	0.00002	0.0232	—
item effect	100	0.00000	0.0068	—

Table 5: What the measure reports when an effect is planted at exactly zero, by observations per item and level, pooled over every other axis. The interaction is flagged well below the nominal 5%; the class effect is not cross-denoised and carries a floor that shrinks with sample size.

The chance level for overlap. With the item and class effects planted orthogonal, the measured overlap tracks r/d closely and sits a little below it (Table 6). This is the number Experiment 2’s overlaps are read against: at $d = 2048$ a genuinely orthogonal pair reads 0.00102, and the values reported in Section 3.3 are two to seven times that.

width d	measured	analytic chance
8	0.19137	0.25000
16	0.10758	0.12500
32	0.05613	0.06250
128	0.01813	0.02344
768	0.00290	0.00391
1024	0.00228	0.00293
2048	0.00102	0.00146

Table 6: Measured item-class overlap when the two effects are planted orthogonal, at 24 observations per item and level. The *chance* column is the analytic level of Appendix A.

C A Construction That Is Not Same-Token

Every construction in the body keys the item on the surface form, so the token is identical at both levels of the distinction and the embedding layer has nothing to distinguish them with. That property is what licenses reading the depth profile as contextualization. Applying the same measure to a distinction that lacks the property shows what its absence looks like.

Number and tense provide that distinction. A plural or a past tense is spelled with an extra subword, so *dog* and *dog-s* are different tokens and the distinction is present in the input before any contextualization occurs.

family	model	feature	layer 0	deepest	ratio
BabyLM	opt-babylm-1.3B-20eps-seed964	Number	0.399	0.254	0.6
BabyLM	opt-babylm-125m-20eps-seed964	Number	0.344	0.221	0.6
BabyLM	opt-babylm-350m-20eps-seed964	Number	0.373	0.219	0.6
Pythia	pythia-1.4b	Number	0.197	0.124	0.6
Pythia	pythia-160m	Number	0.168	0.141	0.8
Pythia	pythia-410m	Number	0.164	0.123	0.7
BabyLM	opt-babylm-1.3B-20eps-seed964	Tense	0.391	0.152	0.4
BabyLM	opt-babylm-125m-20eps-seed964	Tense	0.305	0.137	0.4
BabyLM	opt-babylm-350m-20eps-seed964	Tense	0.360	0.138	0.4
Pythia	pythia-1.4b	Tense	0.185	0.082	0.4
Pythia	pythia-160m	Tense	0.146	0.084	0.6
Pythia	pythia-410m	Tense	0.154	0.084	0.5

Table 7: Number and tense, which are marked by an affix and therefore read at a different subword at the two levels. Unlike every construction in the body, the interaction is large at layer 0 and does not grow, which is what a distinction visible in the input looks like.

The profile inverts (Table 7). Where the same-token constructions begin near zero at layer 0 and grow by one to two orders of magnitude, number and tense begin large and grow barely or not at all. This is what a distinction carried by the input rather than built by the network looks like under this measure, and it is the pattern the body’s constructions were designed to avoid.

It also illustrates why the surface-form keying of part of speech mattered. Keyed on the lemma, part of speech looked like this: a large layer-0 interaction and, for the BabyLM models, no growth. Keyed on the form it looks like role and metaphor instead. The measure was reporting the tokenization correctly in both cases; what changed was which question it was being asked.

D Model Details

family	model	params	width	layers	training data	tokens
BabyLM	opt-babylm-125m	125M	768	12	BabyLM (child-available)	~2B
	opt-babylm-350m	350M	1024	24	BabyLM (child-available)	~2B
	opt-babylm-1.3B	1.3B	2048	24	BabyLM (child-available)	~2B
Pythia	pythia-160m	160M	768	12	the Pile	~300B
	pythia-410m	410M	1024	24	the Pile	~300B
	pythia-1.4b	1.4B	2048	24	the Pile	~300B

Table 8: The six models, paired by width. Training-token counts are approximate; the OPT-BabyLM models train for 20 epochs over a 100M-word corpus, so their effective token exposure is roughly twenty times the corpus size. *Layers* counts the transformer blocks, and the measure is applied to each block’s output plus the embedding layer.

The two families are matched on width rather than on parameter count, which is what the measure is sensitive to: the chance level for overlap is the class subspace rank over the width (Appendix A), so a comparison across models of different width has to be read against different chance levels. Pairing at 768, 1024 and 2048 removes that from the family comparison.

They differ in two ways that matter elsewhere in the paper. The training corpora differ by more than two orders of magnitude in size, which is the manipulation of interest. And they position tokens differently: Pythia uses rotary embeddings, applied inside attention, while OPT learns absolute position embeddings added at the input. Appendix F uses that difference to rule out absolute position as the source of the interaction.

E Deepest-Layer Measurements

Table 9 gives the deepest-layer values behind the curves of Figure 4, together with the chance level each overlap is read against.

distinction	item-class overlap	interaction overlap	class effect	interaction	portable fraction
POS (noun/verb)	0.006 (0.001)	0.191 (0.076)	0.081	0.173	0.31 [0.25, 0.47]
role (subject/object)	0.004 (0.001)	0.131 (0.048)	0.067	0.079	0.46 [0.40, 0.56]
metaphor (literal/figurative)	—	0.227 (0.176)	0.002	0.044	0.07 [0.05, 0.08]

Table 9: Experiment 2, at each model’s deepest layer. The two overlaps are the two ways separability can fail by orientation, each with its own chance level in parentheses: item-class overlap is the fraction of the item effect lying in the class subspace (chance = r_{class}/d), and interaction overlap is the fraction of the interaction lying in the span of the two main effects (chance = r_{margin}/d , much higher because that subspace has rank of order the item count). The portable fraction is the class effect’s share of the class-relevant structure, the class effect divided by the sum of the class effect and the interaction: how much of what distinguishes the two class levels is a direction shared across words rather than specific to each word.

F Positional Embeddings Are Not the Source

One architectural explanation of the interaction has to be ruled out before it can be read as a fact about lexical representation. OPT-BabyLM adds a *learned absolute position embedding* at the input. The two levels of a linguistic distinction are not positionally interchangeable in a sentence, since subjects precede objects and nouns and verbs occupy different slots, so a representation that varies with position varies with the class. If that is what the decomposition is picking up, the interaction would be an artifact of where words sit rather than of how they are represented.

We re-extracted every OPT-BabyLM model with *decoder.embed_positions* zeroed and measured it identically. The ablation was verified per model rather than assumed: with the embedding zeroed, the same token at two different offsets must give identical layer-0 states, and it does, in every model (drift 0.0 to machine precision). Pythia has no such module, so there is nothing to zero there.

The right quantity is the *portable fraction* used in Table 9, the class effect’s share of class-relevant structure. If positions were manufacturing the interaction, removing them should leave a larger share of what distinguishes the two levels in a direction shared across words. Absolute sizes cannot be compared here: zeroing the embeddings in a model that was trained with them is a heavy intervention, and it shrinks every component at once, the class effect included.

model	positions	class	interaction	portable fraction
opt-babylm-125m	zeroed	0.012	0.034	0.264
opt-babylm-125m	intact	0.082	0.105	0.439
opt-babylm-350m	zeroed	0.008	0.023	0.253
opt-babylm-350m	intact	0.080	0.092	0.468
opt-babylm-1.3B	zeroed	0.009	0.035	0.198
opt-babylm-1.3B	intact	0.057	0.115	0.333

Table 10: Trained OPT-BabyLM at the deepest layer, with absolute position embeddings intact and zeroed. If positions produced the interaction, the portable fraction would rise when they are removed. It falls in all three models. Absolute sizes shrink because zeroing the embeddings damages a model trained with them, which is why the ratio rather than the level is the comparison.

Removing the position embeddings makes these representations *less* separable, not more. The portable fraction falls in every model, from 0.439 to 0.264, from 0.468 to 0.253, and from 0.333 to 0.198 (Table 10). Whatever the absolute position embedding contributes, it is contributing to the part of class structure that *is* carried by a shared direction. Take it away and what remains is more word-specific, not less. The interaction is not an artifact of position.

A second line of evidence points the same way and needs no ablation at all. Pythia never had an absolute position embedding to remove, and its portable fractions are 0.253, 0.287 and 0.262, at or below

every intact OPT-BabyLM value. The family without the mechanism shows the phenomenon at least as strongly as the family with it. An explanation resting on absolute position embeddings would have to account for the models that do not have them.

We tested this for part of speech, the construction where the positional worry is sharpest because noun and verb uses of a form occupy systematically different sentence positions. Grammatical role carries the same concern for the same reason, and behaves the same way. Metaphor does not, since literal and figurative uses of a word are not positionally distinguished.